POWER BRAKE BOOSTER (Continued)

The chamber areas forward of the booster diaphragms are exposed to vacuum from the intake manifold. The chamber areas to the rear of the diaphragms, are exposed to normal atmospheric pressure of 101.3 kilopascals (14.7 pounds/square in.).

Brake pedal application causes the primary push rod to open the atmospheric inlet valve. This exposes the area behind the diaphragms to atmospheric pressure. The resulting pressure differential provides the extra apply force for power assist.

The booster check valve, check valve grommet and booster seals are serviceable.

REMOVAL

(1) Remove combination valve and master cylinder.

(2) Disconnect vacuum hose from booster check valve.

(3) Remove retaining clip that secures booster push rod to brake pedal (Fig. 48) and slide the rod off the pin.

(4) Remove four nuts attaching booster to front cowl panel (Fig. 49).

(5) In engine compartment, slide booster studs out of cowl panel, and remove the booster from engine compartment.

(6) Remove dash seal from booster.

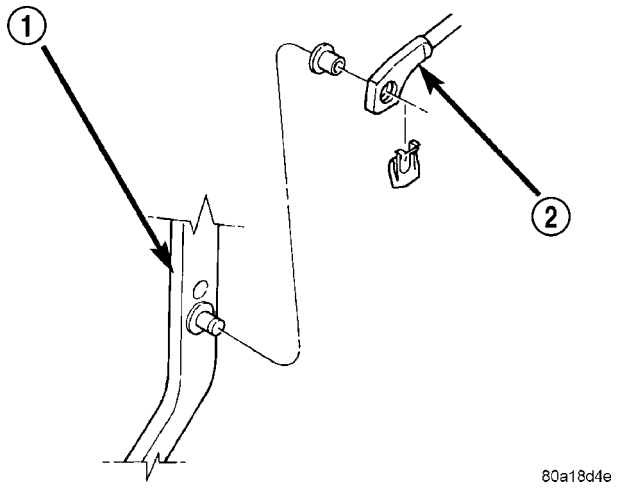

Fig. 48 Push Rod & Clip

1 - BRAKE PEDAL
2 - BOOSTER ROD

INSTALLATION

(1) Clean the booster mounting surface.

(2) Install dash seal on booster.

(3) Align and position booster on the front cowl panel.

(4) In passenger compartment, install nuts that attach booster to dash panel. Tighten nuts just enough to hold booster in place.

(5) Lubricate the pedal pin and bushing with Mopar multi-mileage grease. Then slid the booster push rod onto brake pedal pin and secure with retaining clip.

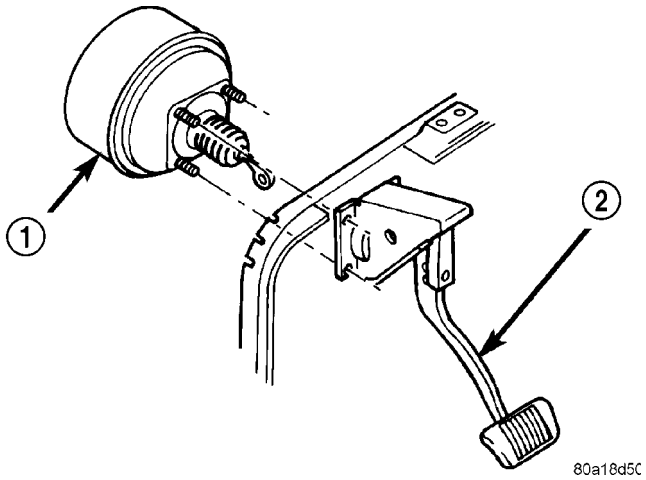

Fig. 49 Booster Mounting Nuts

1 - BOOSTER
2 - BRAKE PEDAL

(6) Tighten booster mounting nuts to 39 N·m (29 ft. lbs.).

(7) Connect vacuum hose to booster check valve.

(8) Install master cylinder and combination valve.

(9) Top off master cylinder fluid level and bleed base brakes.

COMBINATION VALVE

DESCRIPTION

The combination valve contains a pressure differential valve and switch and a rear brake proportioning valve. The valve is not repairable and must be replaced as an assembly if diagnosis indicates this is necessary.

OPERATION

PRESSURE DIFFERENTIAL VALVE

The pressure differential switch is connected to the brake warning light. The switch is actuated by movement of the switch valve. The switch monitors fluid pressure in the separate front/rear brake hydraulic circuits.

A decrease or loss of fluid pressure in either hydraulic circuit will cause the switch valve to shuttle to the low pressure side. Movement of the valve pushes the switch plunger upward. This action closes the switch internal contacts completing the electrical circuit to the red warning light. The switch valve will remain in an actuated position until repairs to the brake system are made.